Refuting the trace-distance case of axiom (T4) of Denis, Lacaille, Martin and Serrano-Ensástiga

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Abstract

We refute the trace-distance case of axiom (T4) of Denis, Lacaille, Martin and Serrano-Ensástiga [1, Sec. III.A.2, Eq. (22) with $p = 1$; (T4) in Sec. II.B, Eqs. (7)–(8)], where "a proof is left open." An explicit SU(2)-covariant channel decreases the order-seven anticoherence of a mixed spin-four state by exactly 95/90552, disproving the universal validity suggested by their reported numerical evidence. The channel is a normalized sum of quadrupole Kraus operations; its covariance and trace preservation follow from Cartesian tensor identities. Both reduced states are diagonal, so the strict inequality reduces to rational arithmetic. We reconstruct an existing spin-four witness and give a self-contained exact counterexample within the full mixed-state domain of the proposed axiom.

1 The question and the witness

Immediately after Proposition 2 in Sec. III.A.2 of arXiv:2605.29436v1, Denis, Lacaille, Martin and Serrano-Ensástiga [1] report numerical evidence for the trace-distance case of (T4) and state that "a proof is left open." The proposed monotonicity is $\mathcal{A}_t^{T,d_1}(\Phi(\rho)) \geq \mathcal{A}_t^{T,d_1}(\rho)$ for every SU(2)-covariant completely positive trace-preserving map Φ (Definition 2, Sec. II.B, Eqs. (7)–(8)). We refute it for a mixed state with $j = 4$ and $t = 7$. The mixed-state definition permits $1 \leq t \leq N = 2j$; the usual restriction $t \leq N/2$ when discussing pure states does not restrict the mixed-state axiom. Pure-state measures are treated in [2]; general equivariant spin channels are studied in [3]. The calculation below uses no classification theorem.

Identify the spin-$N/2$ space with the symmetric subspace of N qubits, using the normalized Dicke basis $\{|D_N^r\rangle : 0 \leq r \leq N\}$, where r counts qubits in state 1. Let ρ_t denote the ordinary partial trace over $N - t$ qubits, expressed in its normalized Dicke basis. With the source's unhalved trace norm, define

$$\mathcal{A}_t^{T,d_1}(\rho) = 1 - \frac{\|\rho_t - \mathbf{1}/(t+1)\|_1}{K_t}, \qquad K_t = \max_{\sigma \geq 0,\, \mathrm{Tr}\,\sigma = 1} \|\sigma - \mathbf{1}/(t+1)\|_1 = \frac{2t}{t+1}. \tag{1}$$

For the last equality, diagonalize σ and express its eigenvalue vector as a convex combination of the vertices of the probability simplex. Convexity of the ℓ^1 norm bounds the distance by its value at a vertex, $2t/(t+1)$, and every vertex attains this value. Including a factor $1/2$ in both distance and K_t leaves the measure unchanged.

The channel acts on the nine-dimensional irreducible spin-four space; the eight-qubit embedding defines the reductions, not a tensor-product noise model. Use the order $r = 0, \dots, 8$, so $m = 4 - r$. Put

$$\begin{aligned} J_z|r\rangle &= (4-r)|r\rangle, & J_+|r\rangle &= \sqrt{r(9-r)}|r-1\rangle, \\ J_- &= J_+^\dagger, & J_x &= (J_+ + J_-)/2, \\ J_y &= (J_+ - J_-)/(2i). \end{aligned} \tag{2}$$

Set $Q_{ab} = \frac{1}{2}(J_aJ_b + J_bJ_a) - \frac{20}{3}\delta_{ab}\mathbf{1}$ for $a, b \in \{x, y, z\}$, and define

$$\Phi(X) = \frac{3}{770}\sum_{a,b} Q_{ab}XQ_{ab}, \qquad \rho = \frac{1}{1764}\operatorname{diag}(209, 92, 560, 0, 441, 0, 294, 168, 0). \tag{3}$$

Theorem 1. *The map Φ in (3) is completely positive, trace preserving and SU(2)-covariant. The state ρ is a mixed density operator, and*

$$\mathcal{A}_7^{T,d_1}(\Phi(\rho)) - \mathcal{A}_7^{T,d_1}(\rho) = -\frac{95}{90552} < 0.$$

Thus this candidate fails (T4) on the full mixed-state domain.

2 An admissible covariant channel

The J_a are Hermitian, satisfy $[J_a, J_b] = i\sum_c \epsilon_{abc}J_c$, and have Casimir $\sum_a J_a^2 = C\mathbf{1}$ with $C = 20$. Consequently every Q_{ab} is Hermitian and $\sum_a Q_{aa} = 0$. The Kraus operators $\sqrt{3/770}\,Q_{ab}$ give complete positivity. We verify their normalization rather than assuming it from covariance.

Lemma 1. *For spin Casimir C, set $S_{ab} = (J_aJ_b + J_bJ_a)/2$ and $Q_{ab} = S_{ab} - (C/3)\delta_{ab}\mathbf{1}$. Then $\sum_{a,b} Q_{ab}^2 = C(4C-3)\mathbf{1}/6$.*

Proof. The commutator relations give $\sum_a [J_a, [J_a, J_b]] = 2J_b$. Expanding the double commutator yields $\sum_a J_aJ_bJ_a = (C-1)J_b$. For $S_{ab} = (J_aJ_b + J_bJ_a)/2$, expansion of the four products gives

$$\sum_{a,b} S_{ab}^2 = \tfrac{1}{2}\sum_{a,b}(J_aJ_bJ_aJ_b + J_aJ_b^2J_a) = \tfrac{1}{2}\big(C(C-1) + C^2\big)\mathbf{1}.$$

Since $\sum_a S_{aa} = C\mathbf{1}$, subtracting the scalar trace part gives $\sum Q_{ab}^2 = (C^2 - C/2 - C^2/3)\mathbf{1}$, as asserted. For $C = 20$ this is $(770/3)\mathbf{1}$; hence the Kraus sum is the identity and Φ is trace preserving. It is also unital. □

For any spin-four rotation U, there is a real orthogonal matrix R with $UJ_aU^\dagger = \sum_c R_{ac}J_c$. Therefore

$$UQ_{ab}U^\dagger = \sum_{c,d} R_{ac}R_{bd}Q_{cd}.$$

Substitution into $U\Phi(X)U^\dagger$ and summation using $\sum_a R_{ac}R_{ae} = \delta_{ce}$ gives $U\Phi(X)U^\dagger = \Phi(UXU^\dagger)$ for every matrix X. This proves covariance for every SU(2) rotation, not just a finite sample.

3 Exact output and partial traces

For an explicit sparse evaluation of (3), set $c = \sqrt{3/3080}$. The same channel has the following five Kraus operators:

$$\begin{aligned} L_0 &= (3J_z^2 - 20\mathbf{1})/\sqrt{1540}, \\ L_{+1} &= -c(J_zJ_+ + J_+J_z), & L_{-1} &= c(J_zJ_- + J_-J_z), \\ L_{+2} &= cJ_+^2, & L_{-2} &= cJ_-^2. \end{aligned} \tag{4}$$

To check the equality of channels, use $Q_{xx} + Q_{yy} = -Q_{zz}$ and write $J_zJ_\pm + J_\pm J_z = 2(Q_{xz} \pm iQ_{yz})$ and $J_\pm^2 = Q_{xx} - Q_{yy} \pm 2iQ_{xy}$. For Hermitian A, B, the cancellation identity is

$$(A + iB)X(A - iB) + (A - iB)X(A + iB) = 2AXA + 2BXB.$$

Apply it with $(A, B) = (Q_{xz}, Q_{yz})$ and $(A, B) = (Q_{xx} - Q_{yy}, 2Q_{xy})$. Also, with $D = Q_{xx} - Q_{yy}$,

$$Q_{xx}XQ_{xx} + Q_{yy}XQ_{yy} = \tfrac{1}{2}(DXD + Q_{zz}XQ_{zz}).$$

The remaining coefficients of $Q_{zz}XQ_{zz}$, $(Q_{xx} - Q_{yy})X(Q_{xx} - Q_{yy})$, and each off-diagonal $Q_{ab}XQ_{ab}$ are respectively 9/1540, 3/1540 and 3/385, exactly those in (3).

For completeness, each input column of (4) has just one possible output. At magnetization $m = 4 - r$, its squared amplitudes and output shifts are

Operator	Output index	Squared amplitude
L_0	r	$(3m^2 - 20)^2/1540$
L_{+1}	$r - 1$	$3(2m + 1)^2[20 - m(m + 1)]/3080$
L_{-1}	$r + 1$	$3(2m - 1)^2[20 - m(m - 1)]/3080$
L_{+2}	$r - 2$	$3[20 - m(m + 1)][20 - (m + 1)(m + 2)]/3080$
L_{-2}	$r + 2$	$3[20 - m(m - 1)][20 - (m - 1)(m - 2)]/3080$

Out-of-range amplitudes vanish. Applying these weights to the probabilities in (3) gives

$$\Phi(\rho) = \frac{1}{194040}\,\mathrm{diag}(22288, 30100, 25528, 13149, 45540, 10584, 24654, 11613, 10584). \tag{5}$$

The input entries are nonnegative and sum to one, with more than one nonzero entry, proving that ρ is a mixed density operator.

The normalized Dicke splitting identity is

$$|D_N^r\rangle = \sum_k \sqrt{\frac{\binom{t}{k}\binom{N-t}{r-k}}{\binom{N}{r}}}\,|D_t^k\rangle \otimes |D_{N-t}^{r-k}\rangle.$$

It follows by grouping computational-basis strings by their first-block weight; the coefficient counts $\binom{t}{k}\binom{N-t}{r-k}$ strings out of $\binom{N}{r}$. Thus the reduction of $\mathrm{diag}(p_r)$ is diagonal with entries

$$p_k^{(t)} = \sum_r p_r \frac{\binom{t}{k}\binom{N-t}{r-k}}{\binom{N}{r}},$$

taking binomial coefficients outside their range as zero. In particular, for $N = 8, t = 7$,

$$p_k^{(7)} = \frac{8 - k}{8}p_k + \frac{k + 1}{8}p_{k+1} \quad (0 \le k \le 7). \tag{6}$$

Applying this formula to the input and output gives

$$\rho_7 = \frac{1}{168}\,\mathrm{diag}(21, 21, 40, 21, 21, 21, 21, 2),$$
$$[\Phi(\rho)]_7 = \frac{1}{517440}\,\mathrm{diag}(69468, 87252, 64205, 82635, 78360, 59892, 43533, 32095). \tag{7}$$

Since $\mathbf{1}/8$ corresponds to $21/168$ or $64680/517440$ in these rows, respectively, summing absolute deviations gives

$$\|\rho_7 - \mathbf{1}/8\|_1 = \frac{38}{168} = \frac{19}{84},$$
$$\|[\Phi(\rho)]_7 - \mathbf{1}/8\|_1 = \frac{117990}{517440} = \frac{3933}{17248}.$$

Their difference is $95/51744$. Dividing by $K_7 = 7/4$ in (1) proves Theorem 1.

4 Consequences and sibling candidates

Mechanism. For a convex mixture of rotations, the partial trace intertwines each collective rotation with its action on the retained qubits. Convexity and unitary invariance of the trace norm then force the reduced distance to decrease. The witness consequently lies outside that subclass of covariant channels. Covariance on the original spin sector alone does not supply a trace-norm contraction on the reduced sector.

At the full order $t = N$, by contrast, trace-distance (T4) always holds. Indeed covariance makes $\Phi(\mathbf{1}/(N+1))$ invariant, hence equal to $\mathbf{1}/(N+1)$ by irreducibility and trace preservation. For Hermitian X, write $X = X_+ - X_-$ with $X_\pm \geq 0$. Positivity and trace preservation give $\|\Phi(X)\|_1 \leq \mathrm{Tr}\,\Phi(X_+) + \mathrm{Tr}\,\Phi(X_-) = \|X\|_1$. Apply this with $X = \rho - \mathbf{1}/(N+1)$ in (1).

The convex-roof construction. Proposition 6 of [1, Sec. III.B.2] assumes that the associated total quantity satisfies the total-measure axioms. Theorem 1 disproves that hypothesis for d_1 on the full mixed-state domain. Consequently that proposition cannot be applied generally to d_1 as a valid total–quantum pair. This is a failure of its hypothesis, not a refutation of the conditional proposition or a separate disproof of the quantum-side axioms.

Fidelity and other orders. The fidelity candidate in [1, Sec. III.A.3, Eq. (26)] is

$$\mathcal{A}_t^{T,F}(\sigma) = \frac{(\mathrm{Tr}\,\sqrt{\sigma_t})^2 - 1}{t}.$$

Its (T4) remains open in that source, including the restatement in Proposition 7 (Sec. III.B.3). Our witness does not refute it.

Proposition 1. *For the state and channel in (3), the fidelity candidate strictly increases at every $t = 1, \ldots, 8$. The trace-distance candidate strictly increases for $t = 1, \ldots, 6, 8$ and decreases for $t = 7$.*

Proof. Table 1 gives the changes, always output minus input.

Here is a finite exact certificate procedure for the table. Compute every diagonal reduction by the binomial formula above. The trace column is then rational arithmetic. For a reduced probability p, set $s = 10^{15}$ and $n(p) = \lfloor\sqrt{\lfloor s^2 p\rfloor}\rfloor$. Integer arithmetic gives $n(p)^2 \leq s^2 p < (n(p)+1)^2$, so

t	$\Delta\mathcal{A}_t^{T,d_1}$ (exact)	$10^6\Delta\mathcal{A}_t^{T,F}$ (lower bound)
1	19/784	3663
2	4275/120736	8574
3	2831/75460	13085
4	4959/241472	18296
5	741/75460	26268
6	19/2464	38910
7	−95/90552	62258
8	3889/17248	365375

Table 1: Exact trace changes and certified rational fidelity bounds for this witness.

$n(p)/s \le \sqrt{p} < (n(p)+1)/s$. Sum these lower and upper bounds to obtain $L_\sigma \le \mathrm{Tr}\sqrt{\sigma_t} < U_\sigma$ for each of the input and output. Both sums are nonnegative; therefore

$$\Delta\mathcal{A}_t^{T,F} \ge \frac{L_{\Phi(\rho)}^2 - U_\rho^2}{t}.$$

Substitution yields a rational bound no smaller than each displayed integer divided by 10^6: the displayed integer is the floor of 10^6 times this certified bound, not a rounded estimate of the true change. The accompanying `siblings.py` and `siblings.json` supply the reductions and exact bounds for all eight rows; no floating-point sign test is used. □

Remaining questions. The universal fidelity-based (T4) is not settled here. Nor does this calculation settle trace-distance (T4) for mixed states restricted to $t \le N/2$, or trace-distance (T4) on pure inputs, in particular in the usual range $t \le N/2$. For arbitrary mixed-state witnesses with $1 \le t \le 2j$, the minimum spin admitting a violation and then the minimum order at that spin are not determined. The existence and corresponding minimum spin/order for pure-input violations with $t \le j$ are separate questions. The proved purity and Hilbert–Schmidt constructions in Propositions 1–2 of [1] are not contradicted by this result.

Attribution. The state and channel are reconstructed from the pinned upstream Lean source, where the declaration `result` states the failure of the mixed-state claim. The argument here supplies a direct Cartesian-tensor proof and exact reduced-state calculation of that witness, rather than a different example.

References

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